



All watched over by machines of loving grace—AI and control, symbolic violence, and the hyperreal

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Abstract

This paper examines the fundamental transformation of social reality through AI systems by synthesizing three critical theoretical frameworks: Baudrillard’s hyperreality, Bourdieu’s symbolic violence, and Deleuze’s societies of control. Moving beyond conventional analyses of AI as technological advancement, it introduces the concept of “meta-simulacra” to theorize how AI systems transcend mere simulation to actively construct reality through recursive feedback loops between algorithmic predictions and human behavior. Through analysis of contemporary technologies—from predictive policing to social credit systems—the research identifies four interconnected mechanisms of algorithmic power: First, as meta-simulacra, AI systems generate hyperreal environments where simulations precede and determine social reality rather than represent it. Second, through algorithmic symbolic violence, these systems naturalize their judgments while concealing their constructed nature, transforming algorithmic classifications into seemingly objective social truths. Third, they implement continuous modulation, where control operates through constant adjustment of possibilities rather than disciplinary constraints. Fourth, they create algorithmic nexus points—critical intersections where information flows, control mechanisms, and reality construction converge to generate new forms of power and control mechanisms. This theoretical synthesis reveals how AI systems produce a novel form of subjectivity—the “modulated self”—where individuals become individuals, simultaneously decomposed into data points and recomposed through algorithmic processes in a control society. The paper demonstrates that this transformation represents not merely a technological evolution but a fundamental shift in how reality is constructed, power is exercised, and human experience is shaped.

Keywords Artificial intelligence · Simulation · Reality construction · Algorithmic power · Symbolic violence

1 Introduction

(...)
“I like to think
(it has to be!)
of a cybernetic ecology
where we are free of our labors
and joined back to nature,
returned to our mammal
brothers and sisters,
and **all watched over**
by machines of loving grace.”
—Richard Brautigan (1967)

The emergence of artificial intelligence (AI) marks not merely a technological revolution but a fundamental transformation in how reality is constructed, power is exercised, and human experience is shaped. While the technological advancement of AI unfolds within postmodernity’s framework of dissolving grand narratives and destabilizing traditional structures (Lyotard 1984, p. 3), its implications extend far beyond mere computational capability. AI systems have become active agents in constructing and maintaining social reality itself, raising fundamental questions about the nature of human experience and agency in an algorithmically mediated world.

This paper examines a central question:

How do AI systems, through their simultaneous roles as generators of hyperreality, instruments of symbolic violence, and mechanisms of continuous control, transform the fundamental nature of human experience and social reality?

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This investigation emerges from the observation that AI systems have evolved beyond tools for processing information to become architects of social reality itself. The analysis synthesizes three theoretical frameworks that illuminate distinct aspects of AI's transformative power: Baudrillard's concept of hyperreality, Bourdieu's theory of symbolic violence, and Deleuze's analysis of control societies. This theoretical triangulation reveals three key mechanisms through which AI systems operate:

First, as meta-simulacra, AI generating hyperreal environments that precede and shape human experience rather than merely representing it. Unlike traditional simulacra, these systems establish recursive feedback loops where AI-generated realities inform human actions, which then generate new data for further reality construction.

Second, through symbolic violence, while maintaining an illusion of neutrality. Despite their apparent objectivity, AI systems perpetuate and reinforce power structures through the subtle imposition of categories and decision-making processes that are presented as universal but remain deeply embedded in specific social contexts.

Third, via continuous modulation, as identified by Deleuze. Rather than operating through discrete disciplinary mechanisms, AI systems establish power through constant adjustment of behavior and perception, creating an algorithmic dispositif that shapes social reality.

Through analysis of contemporary examples, this paper demonstrates how these theoretical frameworks illuminate the mechanisms through which AI systems transform social reality. Moving beyond both techno-optimistic and dystopian narratives, the analysis examines how AI systems simultaneously enable new forms of human capability while constraining human agency in profound ways.

2 Hyperreality and symbolic power

Jean Baudrillard's concept of hyperreality provides a critical framework for understanding the nature of reality in a world increasingly mediated by artificial intelligence. Hyperreality, as Baudrillard (1981, p.1) posits, is a condition in which the distinction between reality and simulation collapses, resulting in a new form of experiential truth that supersedes traditional notions of the real. This concept is inextricably linked to Baudrillard's theory of simulacra and simulation, wherein signs and symbols increasingly replace and precede the real, ultimately generating a reality of their own. Central to Baudrillard's thesis is the notion of the "precession of simulacra", a process by which representations precede and determine the real. In this paradigm, simulations are not mere imitations of reality but become constitutive of reality itself. Baudrillard delineates four stages in the evolution of the

image: reflection of basic reality, masking and perversion of basic reality, masking the absence of basic reality, and pure simulacrum bearing no relation to any reality. This progression elucidates the gradual detachment of signs from their referents, culminating in a hyperreal condition where the simulation becomes more real than reality itself (Baudrillard 1988, p.166).

As Baudrillard eloquently states:

"Simulation is no longer that of a territory, a referential being, or a substance. It is the generation by models of a real without origin or reality: a hyperreal. The territory no longer precedes the map, nor does it survive it. It is nevertheless the map that precedes the territory." (Baudrillard 1981, p.1).

In the context of AI, this concept takes on new significance. AI systems, with their capacity to generate, process, and manipulate vast quantities of data, serve as powerful engines of simulation, often creating representations that are indistinguishable from or preferable to traditional notions of reality. Machine learning algorithms, for instance, operate on datasets that are themselves abstractions or simulations of reality. The increasing use of synthetic data—artificially generated information designed to mimic the statistical properties of real-world data—further blurs the line between the authentic and the simulated. (Hao et al. 2024, p.2) These synthetic datasets, created by AI systems to train other AI systems, represent a higher order of simulation, echoing Baudrillard's concept of simulacra that precede and shape reality. The deployment of AI trained on such synthetic data introduces a layer of hyperreality into decision-making processes, where predictions and actions are based on simulations of simulations, potentially diverging further from ground truth realities. (Zhang et al. 2025, p. 15) The phenomenon of "deep fakes" and AI-generated content, like Virtual Reality or Augmented Reality (VR/AR), further exemplifies the hyperreal condition in the digital age. These technologies produce simulations that are increasingly indistinguishable from reality, challenging traditional epistemological frameworks and notions of truth. (Chesney & Citron 2019, p. 1756–1758) In this hyperreal environment, AI-enhanced and generated content becomes the new standard against which reality is judged, creating a feedback loop where artificial constructs shape user desires, which in turn inform future AI creations. Baudrillard's concept of the "implosion of meaning" is particularly relevant in the age of AI-driven information systems.

As he notes, "(...) *we think that information produces meaning, the opposite occurs. Information devours its own content. It devours communication and the social. (...) Rather than creating communication, it exhausts itself in the act of staging communication. Rather than producing meaning, it exhausts itself in the staging of meaning. A gigantic*

process of simulation that is very familiar." (Baudrillard 1981, p. 80–83).

This implosion occurs when the proliferation of AI-generated information leads not to enlightenment but to a collapse of meaning, where the excess of signification paradoxically results in a loss of signified content. The application of Baudrillard's hyperreality to AI systems reveals profound implications for society. Predictive policing algorithms, for example, operate on historical crime data (itself a representation) to generate simulations of future criminal activity that subsequently inform real-world policing strategies, thus bringing the simulation into reality (Lally 2022, p.1). This exemplifies Baudrillard's precession of simulacra, where the model precedes and shapes the real.

In conclusion, Baudrillard's concept of hyperreality provides a powerful lens through which to examine the societal impact of AI. As AI systems increasingly mediate our perception and interaction with the world, they create a hyperreal environment where simulations not only reflect but actively construct reality. This challenges fundamental notions of truth, authenticity, and human agency, necessitating a critical reevaluation of our relationship with AI technologies and the realities they generate.

2.1 Bourdieu's symbolic violence

Pierre Bourdieu's concept of symbolic violence provides a nuanced framework for understanding how power operates through subtle, often invisible mechanisms of domination. This theoretical lens offers profound insights into the societal impact of artificial intelligence, revealing how AI systems can perpetuate and exacerbate existing power structures while maintaining an illusion of neutrality and objectivity. Symbolic violence can now be defined as a generally unperceived form of violence. This is in contrast to systems in which force is needed to maintain social hierarchy and is thus an effective and efficient form of domination in that members of the dominant classes need exert little energy to maintain their dominance. (Grenfell, 2014, p.184) It is the imposition of systems of symbolism and meaning upon groups or classes in such a way that they are experienced as legitimate. This form of violence is "symbolic" in that it is enacted through cognition, misrecognition, and complicity rather than physical force.

Central to Bourdieu's theory is the concept of "habitus," a system of dispositions that individuals acquire through socialization, which shapes their perceptions, appreciations, and actions. The habitus, being a product of social conditions, tends to reproduce those conditions, thus perpetuating existing power structures (Bourdieu 1977, p.78; Power 1999, p.49). In the context of AI, we can conceptualize a "digital habitus"—the internalized norms and practices that shape how individuals interact with and perceive AI systems.

Bourdieu also introduces the notion of "doxa," the taken-for-granted, unquestioned truths that underpin social fields. As Bourdieu states,

"Doxa is the point at which the socially constructed world appears as natural. It is the set of fundamental beliefs which does not even need to be asserted in the form of an explicit, dogmatic assertion." (Bourdieu 1977, p.164).

In the realm of AI, the belief in algorithmic objectivity and efficiency can become doxic, making the use of AI systems seem inevitable and their decisions unquestionable. Furthermore, Bourdieu's concept of "cultural capital"—non-financial social assets that promote social mobility, such as education, intellect, and aesthetic preferences—is crucial in understanding the mechanisms of symbolic violence. The unequal distribution of cultural capital serves as a means of reproducing social hierarchies through symbolic means (Grenfell, 2014, p. 101). In the age of AI, technological literacy and access to AI tools can be seen as new forms of cultural capital, potentially exacerbating existing inequalities.

2.2 Dimensions of power and domination in the digital age

The application of Bourdieu's theory of symbolic violence to AI and algorithmic decision-making systems reveals new dimensions of power and domination in the digital age. AI systems, often perceived as neutral and objective, can be understood as instruments of symbolic violence, imposing systems of meaning and classification that are experienced as legitimate despite their potential for perpetuating or exacerbating social inequalities, leading to what Bursell & Roumbanis call a process of meta-algorithmic judgement (Bursell and Roumbanis 2024, p.4).

Consider, for example, an AI-driven hiring system employed by a major corporation. This system, trained on historical hiring data, uses machine learning algorithms to screen resumes and predict candidate success. (Bursell and Roumbanis 2024, p.1) On the surface, it appears to be an objective, efficient tool for selecting the most qualified candidates. However, this system embodies several aspects of Bourdieu's symbolic violence:

1. The AI system, trained on past hiring decisions, internalizes and reproduces the corporation's existing preferences and biases. It develops a 'digital habitus' that shapes its perceptions and decisions, perpetuating historical hiring patterns.
2. The use of AI in hiring becomes an unquestioned truth within the corporate world. The belief in AI's objectivity and efficiency becomes doxic, making its use seem inevitable and its decisions unquestionable.

3. The AI system values certain forms of cultural capital, such as degrees from prestigious universities or specific linguistic patterns in resumes. This unequal valuation of cultural markers serves to reproduce existing social hierarchies.
4. The AI system imposes its classifications and judgments upon job applicants in a way that is experienced as legitimate. Candidates internalize the system's decisions, potentially altering their behavior or self-perception to align with the AI's preferences.

This AI hiring system thus becomes an instrument of symbolic violence, subtly enforcing and perpetuating existing power structures while maintaining an illusion of neutrality and objectivity. Its decisions, shaped by historical biases and doxic beliefs about merit and qualification, disproportionately affect marginalized groups. Yet, because the system is perceived as objective and its inner workings are opaque, its role in reproducing social inequalities remains largely unrecognized and unchallenged. (Bursell & Roumbanis, Bursell and Roumbanis 2024, p. 15) In the broader context of machine learning algorithms, the selection and structuring of training data can be seen as a form of imposing categories and classifications, akin to Bourdieu's notion of symbolic power. These algorithms, trained on historically biased data, may reproduce and amplify existing social inequalities while presenting their outputs as objective and neutral (Noble 2018, p. 7). When AI systems make decisions that disproportionately affect marginalized groups – for instance, in credit scoring, hiring processes, or criminal justice – they are engaging in a form of symbolic violence by imposing and legitimizing systems of classification that perpetuate social hierarchies (Eubanks 2018, p. 178). Moreover, the opacity of many AI systems, often referred to as the “black box” problem, can be understood as a manifestation of doxa in the digital realm. The inner workings of these systems remain unquestioned and taken for granted, their decisions accepted as inevitable and legitimate, thus exemplifying the operation of symbolic violence (Pasquale 2015, p.189–218).

In conclusion, Bourdieu's concept of symbolic violence provides a powerful framework for critically examining the societal impact of AI. It reveals how AI systems, far from being neutral tools, can become instruments of power that subtly reinforce and legitimize existing social hierarchies. By recognizing the symbolic violence inherent in AI systems, we can begin to address the hidden power dynamics at play and work towards more equitable and just applications of this transformative technology.

2.3 AI at the intersection of hyperreality and symbolic violence

The synthesis of Jean Baudrillard's hyperreality and Pierre Bourdieu's symbolic violence offers a potent framework for analyzing AI's societal impact. This intersection reveals how AI systems not only shape our perception of reality but also impose systems of meaning that perpetuate power structures. AI-driven simulations generate hyperreal representations that precede and shape reality, becoming instruments of symbolic violence. These systems impose categories and decision-making processes that appear objective and legitimate, despite being rooted in existing social hierarchies. Baudrillard (1981) argues, “*The simulacrum is never what hides the truth—it is truth that hides the fact that there is none.*” (Baudrillard 1981, p.1) In AI context, this suggests that AI-created simulations become the new reality against which all else is measured.

Consider AI-powered social media algorithms. These create hyperreal information environments while engaging in symbolic violence by shaping users' perceptions and reinforcing power dynamics. Users often willingly participate in these systems, unaware of how their perceptions are being shaped. This synthesis also reveals new forms of cultural capital in the digital age. Proficiency with AI technologies becomes valuable cultural capital, potentially exacerbating social inequalities. Those possessing this capital gain advantages across various fields, while others may face marginalization. This aligns with Bourdieu's observation that these new forms of capital build upon existing social structures and information flows. (Grenfell 2014, p.159) From this synthesis emerges “algorithmic doxa”, unquestioned beliefs embedded within AI systems. Shaped by AI-generated hyperreal simulations, these beliefs become instruments of symbolic violence, imposing systems of meaning experienced as legitimate and inevitable. Bourdieu (1977) states in this context that, “*The most successful ideological effects are those which have no need of words, and ask no more than complicitous silence.*” (Bourdieu 1977, p.188) Algorithmic doxa operates in this silent, pervasive manner. These layers of abstraction create hyperreal worlds increasingly detached from any original reference. Baudrillard warns in this context:

“It is no longer a question of imitation, nor duplication, nor even parody. It is a question of substituting the signs of the real for the real.” (Baudrillard 1981, p.2).

As these systems become more sophisticated, they have the potential to reshape the foundations of social reality and power dynamics.

3 AI as meta-simulacra

While Baudrillard's concept of simulacra describes representations that eventually come to replace reality, AI systems represent a novel evolution of this phenomenon, what we might term meta-simulacra. Unlike traditional simulacra, which function as passive representations eventually standing in for the real, AI as meta-simulacra actively constructs and shapes reality through dynamic processes of interpretation, generation, and manipulation—generating simulations of simulations. (Shanahan et al. 2023, p. 495) This distinction is crucial: where traditional simulacra operate in a largely unidirectional manner, replacing reality with its representation, meta-simulacra establish recursive feedback loops in which AI-generated realities inform human actions, which in turn generate new data that feeds back into AI systems. These systems don't merely represent or replace reality; they actively participate in its ongoing construction, wielding symbolic power that fundamentally alters human perception, cognition, and decision-making across multiple domains simultaneously. (Shanahan et al. 2023, p. 493) Perhaps most significantly, while traditional simulacra may maintain some reference to an original "real," meta-simulacra can generate content and shape realities that have no original referent, effectively transcending the traditional relationship between copy and original that characterized earlier forms of simulation.

The emergence of generative AI models, such as GPT-4o, O3 or even systems of higher machine intelligence, further blurs the boundaries between simulation and reality construction. These systems produce (super-)human-like text, images, videos, and code, generating content increasingly indistinguishable from human-produced artifacts. (Zhang et al. 2025, p. 3; Grace et al. 2024, p.3).

Shanahan et al. describe this effect in their work *Role play with large language models* as the following:

“Likewise, a simulacrum can play the role of a character with full agency, one that does not merely act but acts for itself. Insofar as a dialogue agent's role play can have a real effect on the world, either through the user or through web-based tools such as email, the distinction between an agent that merely role-plays acting for itself, and one that genuinely acts for itself starts to look a little moot, and this has implications for trustworthiness, reliability and safety.” (Shanahan et al. 2023, p.496).

As AI-generated outputs proliferate, they constitute a significant portion of the information ecosystem, actively shaping discourse, aesthetics, and knowledge production. (Doshi & Hauser 2024, p.4) Baudrillard (1981, 1987,

1988) insightfully describes in his works the emergence of a social condition in which technological simulations increasingly replace authentic experiences. In this emerging reality, what was once considered real becomes indistinguishable from its technological representations, until eventually, the distinction dissolves entirely. Reality, as such, is no longer experienced directly but is mediated and defined by simulations that stand in for it, making the simulation itself the only remaining reference point.

Baudrillard states in this context:

“(…) there is not only an implosion of the message in the medium, there is, in the same movement, the implosion of the medium itself in the real, the implosion of the medium and of the real in a sort of hyper-real nebula, in which even the definition and distinct action of the medium can no longer be determined.” (Baudrillard 1981, p.82).

This underscores the profound implications of AI as meta-simulacra. As AI systems increasingly furnish our reality with simulations, the distinction between the real and the simulated becomes not just blurred, but potentially irrelevant. This view is supported by Shanahan et al. who state: “(…) we can see a dialogue agent as a superposition of simulacra within a multiverse of possible characters.” (Shanahan, 2023, p.493).

The result is a hyperreal environment where AI-generated simulations precede and shape our understanding of reality itself. Each interaction with such systems becomes less about uncovering a stable truth and more about navigating a fluid constellation of possible selves, none of which are anchored in a singular ontological ground. This dynamic not only reshapes perception but also initiates recursive feedback loops, where simulations influence the very realities they purport to represent, thereby reinforcing their own epistemic authority.

3.1 The recursive nature of AI-generated realities

This recursion is already evident in various domains such as algorithmic trading systems which make decisions based on market data while simultaneously influencing market trends, creating a complex interplay between AI predictions and market realities (Kirilenko & Lo 2013, p. 5). The recursive dynamic is leading to an algorithmic co-construction of reality where human and AI influences become increasingly indistinguishable.

Bourdieu's concept of symbolic power is particularly relevant here:

“Symbolic power is a power of constructing reality and, as such, it is a specifically symbolic logic”. (Bourdieu 1991, p.166).

The profound implications of these mechanisms lie in their capacity to naturalize constructed patterns of reasoning. By embedding specific logical structures within technological systems, AI transforms contingent choices into apparently inevitable truths. This process of naturalization operates through the integration of computational logic into broader systems of thought, language, and cultural production. AI systems wield this symbolic power, constructing reality through what is recognized as legitimate or true. Operating within thought, language, and culture, this power is often internalized as “normal”, making it challenging to detect or resist. The pervasive influence of AI in shaping public discourse, from news recommendations to social media trends, has the potential to alter collective perception and social dynamics. The rapid dissemination of information and shaping of narratives by AI systems raises questions about public opinion formation in an age of algorithmic influence (Aral 2020, p.24). Moreover, as shown, the increasing reliance on AI for decision-making in domains such as healthcare or criminal justice raises critical questions about human agency and autonomy. For example, as AI surveillance and military systems continue to gain capability, their growing sophistication poses a clear risk that human judgment will be supplanted by AI-generated recommendations in highly critical domains (Ulbrecht & Egbert, 2024, p. 3).

The implications of AI as meta-simulacra extend beyond immediate perceptual and cognitive effects, potentially reshaping fundamental concepts of identity, truth, and reality. By moving beyond mere simulation to active reality construction, AI systems are thus fundamentally altering the relationship between humans and their informational environment, especially in the area of control.

4 Control and modulation: Deleuze and the algorithmic society

The emergence of artificial intelligence marks not merely a technological evolution but a fundamental transformation in the mechanisms of social control. While previous sections of this paper have examined AI through the lenses of Baudrillard’s hyperreality and Bourdieu’s symbolic violence, this part of the paper now introduces Gilles Deleuze’s (1992) theory of control societies to illuminate how AI systems establish and maintain power through continuous modulation of behavior and identity. This theoretical framework reveals how AI represents not just a new tool of control but a paradigmatic shift in how power operates in contemporary society. In this sense, Deleuze’s framework of modulation converges with Bourdieu’s concept of symbolic power, as algorithmic systems reconfigure the field of social relations by encoding and operationalizing forms of capital—such

as reputation, trust, or visibility—into quantifiable metrics, thereby rendering symbolic violence not only more opaque but also more pervasive and automatic.

Deleuze’s (1992) prescient observation that “*the societies of control operate with machines of a third type, computers*” (Deleuze, 1992, p.6) anticipates the fundamental role of AI in establishing new mechanisms of social control. Unlike disciplinary societies, which operated through distinct sites of confinement (schools, factories, prisons), societies of control function through continuous monitoring and modulation. Consider the transformation of workplace surveillance through AI systems. Traditional factory discipline, characterized by fixed schedules and spatial confinement, gives way to algorithmic management systems that continuously monitor and modulate worker behavior. As Moore and Robinson (2016) demonstrate, platforms like Amazon’s worker tracking systems create a “quantified worker” whose every movement and decision is subject to real-time optimization. (Moore and Robinson 2016, p. 2779) Similarly, gig economy platforms like Uber employ AI systems that dynamically adjust prices, distribute work, and monitor performance, creating what Rosenblat calls “algorithmic management”. (Rosenblat 2018, p. 138–166) Through Zuboff’s (2019, p.14) lens this crisis of institutional enclosure represents not merely a breakdown of traditional disciplinary structures, but their evolution into what she terms “instrumentarian power”, a new form of control that operates through the computational architecture of surveillance capitalism, an interconnected regime of ubiquitous sensors, algorithms, and actuators that render traditional institutional boundaries obsolete.

Zuboff elaborates as the following:

" (...) instrumentarian power works its will through the automated medium of an increasingly ubiquitous computational architecture of ‘smart’ networked devices, things, and spaces." (Zuboff 2019, p.14).

These systems therefore represent not just enhanced surveillance but a new logic of control that operates through continuous adjustment rather than discrete punishment while reducing individuals to adjustable data points through a process of algorithmic decomposition.

4.1 The dividual and digital fragmentation

The concept of the dividual, introduced by Deleuze to characterize the subject of control societies, marks a fundamental shift from individual identity to algorithmic decomposition and is of central relevance to understand the described dynamics. Unlike the individual – understood as an indivisible unit of disciplinary society – the dividual emerges through processes of digital fragmentation and continuous recombination. (Deleuze, 1992, p.5) This transformation

represents not merely a technological development but a fundamental reconceptualization of human subjectivity in the age of AI systems. The transition from individual to dividual occurs through the reduction of human experience to discrete, measurable data points, their continuous algorithmic processing, and subsequent dynamic reconstruction into multiple, context-specific digital identities. This process creates what Deleuze (1992, p.7) terms "numerical bodies", entities that can be endlessly divided, recombined, and modulated according to algorithmic imperatives. The dividual thus exists not as a unified subject but as a series of data-driven assemblages, each corresponding to different algorithmic contexts and purposes.

Deleuze describes this process of decomposition as the following:

"The numerical language of control is made of codes that mark access to information, or reject it. We no longer find ourselves dealing with the mass/individual pair. Individuals have become 'dividuals,' (...)" (Deleuze, 1992, p.5).

As shown, contemporary AI systems now operationalize the dividual through various forms of algorithmic assemblage, constructing behavioral profiles and predictive models that span transaction patterns, communication habits, movement trajectories, and social connections. These assemblages don't simply represent existing identities; they actively construct new forms of digital subjectivity. As Cheney-Lippold (2017) argues, these "measurable types" become self-fulfilling prophecies, shaping behavior through their predictive operations. (Cheney-Lippold 2017, p.249–266) As described, the dividual therefore exists simultaneously across multiple digital platforms and systems, each constructing its own version of the subject. Social media platforms for example similarly fragment and recombine user identities through AI-driven recommendation systems. As Zuboff (2019, p.194) details, platforms like Facebook construct multiple versions of each user's identity based on behavioral data, creating what might be termed "preference dividuals" that shape content delivery and advertising targeting. These algorithmic identities often diverge significantly from users' self-conceived identities, yet they increasingly determine users' information environments and social experiences.

4.2 Modulation and reality construction

Deleuze's concept of continuous modulation provides a crucial theoretical framework for understanding how AI systems construct and exploit the hyperreal reality of the modern dividual through dynamic adjustment rather than fixed impositions. Unlike disciplinary society's rigid molds,

control societies operate through fluid modulations that continuously reshape social reality. As Deleuze articulates:

"Enclosures are molds, distinct castings, but controls are a modulation, like a self-deforming cast that will continuously change from one moment to the other." (Deleuze, 1992, p.4).

This modulation represents a fundamental shift in how power operates: from discrete institutional impositions to continuous algorithmic adjustments. While this process aligns with Baudrillard's concept of hyperreality, it emphasizes the operational control mechanics of reality construction. As described, this continuous modulation manifests across various technological systems. The conceptual ambition of the Chinese Social Credit System represents perhaps the most comprehensive implementation of algorithmic modulation. Unlike traditional disciplinary systems that sought to mold fixed identities, this AI-driven apparatus continuously recombines various data points – financial transactions, social media activity, travel patterns, personal relationships – to generate dynamic "trustworthiness" scores. These scores, as Liang et al. (2018, p.1) argue, function not as fixed classifications but as fluid assemblages that modulate access to services, opportunities, and social status. Importantly, similar forms of algorithmic modulation and reputation scoring are already operational in Western contexts, albeit often in more decentralized or commercially driven forms. For example, ride-sharing platforms like Uber continuously rate drivers and passengers, influencing access to future work or services (Rosenblat 2018, p. 78). Similarly, credit scoring agencies in the United States and Europe assign numerical risk profiles that determine eligibility for loans, housing, and insurance—often based on opaque data practices (Eubanks 2018, p. 12). Social media platforms also implement dynamic recommendation systems that shape visibility, influence, and social engagement through behavioral scoring (Zuboff 2019, p. 194). These examples suggest that the logics of algorithmic modulation, data-driven scoring, and behavioral governance are not unique to authoritarian regimes but are increasingly characteristic of digital societies globally. (Bonini and Tréré, 2024, p.16).

Taken together, Deleuze, Baudrillard, and Bourdieu offer a multidimensional understanding of algorithmic power: Deleuze illuminates the shift toward continuous modulation and the emergence of the dividual as the primary subject of control; Baudrillard reveals how AI systems fabricate hyperreal environments that collapse the boundary between representation and reality; and Bourdieu exposes how these algorithmic structures enact symbolic violence by encoding social hierarchies into seemingly neutral systems of classification. In unison, these theorists show that algorithmic governance reshapes subjectivity, naturalizes inequality, and constructs entire ontologies of the real through the

recursive interplay of modulation, simulation, and symbolic domination.

5 Specters of the hyperreal: modulated subjects, meta-simulacra and nexus points of control

The convergence of Baudrillard’s hyperreality, Bourdieu’s symbolic violence, and Deleuze’s societies of control now provides a powerful theoretical framework for understanding the existential threats posed by AI systems. Rather than examining these dangers through a single lens, their synthesis reveals how AI simultaneously reconstructs reality, modulates behavior, and transforms power relations in contemporary society. Baudrillard’s hyperreality takes on new significance in the context of advanced AI systems, as the process reaches its apex in AI systems that not only simulate but actively construct reality through predictive models and generative capabilities. The “precession of simulacra” becomes literal as AI models as meta-simulacra precede and determine social, economic, and political realities through continuous modulation. This modulation manifests in AI systems that shape decision-making across all social domains, as shown by the examples described. These systems exemplify both Baudrillard’s simulacra and Deleuze’s control mechanisms, creating self-fulfilling prophecies through the continuous adjustment of possibilities.

As Deleuze fittingly notes, this leads to a never-ending process of adjustment via modulation:

“(…) in the societies of control one is never finished with anything (…)” (Deleuze, 1992, p.5).

This modulation is thus perpetual and pervasive. The result is not merely a simulation of reality but its active construction through algorithmic means, where the excess of AI-generated information paradoxically results in the collapse of meaning itself. Consequently, the transformation of surveillance from fixed to fluid forms marks a crucial shift in power relations. Unlike the Benthamian panopticon’s architectural determinism (Bentham 2020, p.29), today’s algorithmic control operates through continuous modulation of behavior and identity. The certainty of modulation replaces the possibility of observation as the primary mechanism of control. Bourdieu’s framework of symbolic violence and capital takes on new dimensions in this context. AI systems become instruments of symbolic violence, where power operates not through fixed dispositions but through continuous modulation of the habitus itself.

As Bourdieu argues:

“The dominated apply categories constructed from the point of view of the dominant to the relations of domi-

nation, thus making them appear natural” (Bourdieu 2001, p.35).

In the AI context, this naturalization occurs through the continuous adjustment of perceptions, preferences, and possibilities. A new form of capital emerges—the ability to influence and navigate systems of algorithmic control. This convergence of hyperreality, continuous control, and symbolic violence suggests a future where power operates through the simultaneous reconstruction of reality, modulation of behavior, and transformation of social structures. The individual becomes the perfect subject of this hyperreal control society, as AI systems simultaneously fragment identity into discrete data points while modulating these fragments through continuous feedback loops. The dividualization of the individual enables the ability to adjust not just behaviors but the very components of digital identity itself, creating subjects who are simultaneously decomposed and recomposed through algorithmic processes.

5.1 Nexus points of control

At the center of this transformation is what shall be called the algorithmic nexus, a concept inspired by Yuval Harari’s work *Nexus* (Harari 2024, p. 279–289). In more general scholarly discourse, a nexus might be understood as a juncture or pivotal linkage that unites discrete elements, functioning as an integrative hub through which multiple components converge and interact in a unified system. The idea of the algorithmic nexus now refers to key points where different technological forces come together as a computational network: the flow of data, systems of control, and the way our reality is shaped by algorithms. At these points of convergence, things like simulation (the creation of digital versions of reality), modulation (the constant tweaking of behavior), and symbolic violence (subtle forms of control that feel natural) all blend together. They no longer function separately but reinforce each other, forming a powerful and invisible infrastructure of influence and control.

To illustrate the concept of the algorithmic nexus consider the following example:

In social media platforms, recommendation engines collect each click, like, or share and integrate them into large-scale data streams that guide real-time content selection. Viewing history and engagement metrics are processed by predictive models (information flows), which then adjust rankings and send notifications to highlight certain videos or posts (control mechanisms). Simultaneously, the user interface, such as the “For You” page or news feed, is designed to present content in ways that reflect specific themes, tones, or social norms (reality construction). Over time, differences in what is promoted or demoted can lead to certain perspectives, topics, or styles appearing more frequently than others.

Through the combined effect of these interconnected processes, users' viewing experiences are continuously shaped by the underlying algorithms. (Galeazzi et al. 2024, p.2) This example now helps to consider how recommendation algorithms function as nexus points: They simultaneously generate hyperreal environments through content curation, implement symbolic violence through the naturalization of algorithmic categorization, and exercise continuous control through behavioral modulation. Yet at these nexus points, these mechanisms don't merely coexist but actively reinforce and amplify each other, creating what can be defined as a convergent power, a form of control that operates through the simultaneous modulation of reality, identity, and behavior. The already introduced Chinese Social Credit System may represent perhaps a comprehensive implementation of this nexus power, where surveillance, reality construction, and behavioral modulation become completely intertwined. The concept of the algorithmic nexus thus reveals how contemporary AI systems create points of power convergence where various control mechanisms become inseparable and self-reinforcing. This helps explain why resistance to algorithmic control becomes increasingly difficult—challenging one aspect of control at a nexus point inevitably means confronting an entire system of interconnected power mechanisms. (Bonini and Treré, 2024, p.14) The perfect subject of this nexus power is now indeed the *dividual*, whose fragmented digital identity becomes not just the target of control but the very medium through which power operates. As AI systems grow more sophisticated, these nexus points multiply and interconnect, creating what constitutes a larger 'nexus network' or a *nexus of nexuses*—a meta-structure of converging power mechanisms that shapes both individual experience and collective reality. (Harari 2024, p.239–248; Bonini and Treré, 2024, p.16) This represents the culmination of what Deleuze foresaw in control societies, what Baudrillard identified in hyperreality, and what Bourdieu analyzed in symbolic violence—a point where these different forms of power converge to create a new and more totalizing form of control.

To illustrate the points made, it is helpful to envision how such a world could look like.

5.2 Modulations: 24 h in the control society

Maya wakes to her personalized AI assistant's voice, calibrated to match her current stress levels based on her sleep data. Her social credit score dropped two points overnight—her smart fridge reported unusual eating patterns, and her streaming service noted she fell asleep during an educational program. The assistant suggests a morning routine optimization: a specific breakfast, a curated news feed, and recommended social interactions to rebuild her score. Her AR lenses activate automatically, overlaying her environment

with personalized prompts and notifications. The kitchen counter displays floating nutritional guides tailored to her genetic profile and current health metrics. Her social feed highlights friends whose interaction would positively impact her algorithmic standing—their profiles marked with compatibility scores and conversation suggestions generated by predictive social models. At work, Maya's performance metrics fluctuate in real-time based on her keyboard patterns, facial expressions, and vocal tonality. The office AI continuously adjusts her workspace environment—lighting, temperature, background music—modulating her productivity of seemingly senseless, but always highly important tasks through imperceptible changes. Her project assignments are automatically redistributed based on her emotional state, physical energy levels, and social dynamics within her team. During lunch, she receives a notification that her "cultural engagement" score needs attention. The AI recommends specific restaurants, suggesting dishes that would simultaneously improve her health metrics and social standing. Her AR lenses highlight other diners whose profiles indicate beneficial networking opportunities, their potential value to her career quantified and displayed in her field of vision. Evening entertainment options are dynamically generated based on her day's performance metrics. The content adapts in real-time to her emotional responses—scenes, music, and narratives shift subtly to maintain optimal engagement. Even her dreams will be analyzed and fed back into the system, refining tomorrow's reality before it begins.

Maya doesn't question this constant modulation. The system's suggestions feel natural, almost like her own thoughts. Her fragmented digital identities across various platforms are continuously recombined and optimized at different nexus points, creating new versions of herself that feel simultaneously foreign and familiar. She is neither fully autonomous nor completely controlled, but perpetually modulated in a hyperreal space where the distinction between her authentic desires and algorithmically generated preferences has become meaningless.

5.3 Coda: our digital and modulated selfs

Maya's daily experience illustrates how AI systems have evolved from mere tools of surveillance into architects of social reality. Her fragmented existence as a *dividual*, decomposed into data points and recomposed through algorithmic processes, embodies what Deleuze envisioned in the shift from disciplinary to control societies. The continuous modulation of her behavior, environment, and identity unfolds so seamlessly that its artificial nature becomes imperceptible. This achieves what Bourdieu would identify as the most insidious form of symbolic violence, in which domination is rendered invisible by appearing natural. In this context, hyperreality does not

manifest through dramatic simulations but through the quiet, relentless recalibration of everyday life. The AI systems that surround Maya do not merely reflect or predict reality. They actively construct it. Her social credit score, wellness metrics, and cultural engagement ratings are not neutral measurements. They become generative forces, producing the very behaviors and outcomes they claim to measure. Maya's story reveals how the convergence of hyperreality, continuous control, and symbolic violence creates a new form of subjectivity. This subject is neither fully autonomous nor entirely determined but exists in a condition of constant modulation. In such a state, the boundary between authentic desire and algorithmic suggestion becomes indistinguishable. She inhabits a hyperreal nebula, a digitally constructed environment in which predictive systems, feedback loops, and symbolic hierarchies blur the contours of reality itself. Within this opaque and immersive atmosphere, the simulated becomes more consequential than the real, and truth is continuously recalibrated by flows of data. This process is captured powerfully in the metaphor of digital human twins; real-time, data-driven replicas that simulate individuals and enable systems to anticipate behavior, automate interventions, and assess identity based on prediction rather than actual actions. Maya is no longer simply shaped by these systems. Her digital twin becomes a simulacral proxy through which decisions are made, risks are evaluated, and futures are constructed. These twins cease to be representations and instead form a new ontological layer of simulation. They are anticipatory models of the self that systems trust more than the embodied person, completing the transition from lived subjectivity to governed simulation. What Maya experiences is not just a technological shift but a profound transformation of human life. It is the rise of the modulated self, a subject continuously adjusted by external systems yet perceiving those adjustments as self-originating and inevitable.

As evident today, the contours of Maya's hyperreal existence are already materializing across Western societies. When smartphones track behavioral biometrics to anticipate mood-based decisions (Hart et al. 2022, p.1), when Netflix and TikTok deploy reinforcement learning to keep users in infinite scroll loops (Afsar et al. 2022, p.1), or when advanced conversational agents like GPT-4 turbocharge customer service, sales, and companionship apps (Wulf and Meierhofer 2024, p.2; De Freitas et al. 2024, p.49), when experimental technologies like brain-machine interfaces begin to materialize in reality (Waisberg et al. 2024, p.1937) and the advent of an Artificial General Intelligence (AGI) may become a reality in the near future (Raman et al. 2025, p.1), we are no longer observing the future, we are living within it. The difference between

Maya's world and ours is no longer a question of speculative fiction, but of scale and saturation.

6 Conclusion

This paper has examined how AI systems fundamentally transform social reality through the synthesis of three theoretical frameworks: Baudrillard's hyperreality, Bourdieu's symbolic violence, and Deleuze's societies of control. Through this analysis, we have revealed how AI operates as meta-simulacra—systems that don't merely simulate but actively construct reality through recursive feedback loops between algorithmic predictions and human behavior.

Four key mechanisms emerge: First, AI systems generate hyperreal environments where simulations precede and shape reality rather than represent it. Second, through "algorithmic symbolic violence", these systems naturalize their judgments while concealing their constructed nature. Third, they implement continuous modulation, where control operates through constant adjustment of possibilities rather than fixed constraints. Fourth, they create algorithmic nexus points—critical intersections where information flows, control mechanisms, and reality construction converge to generate new forms of power through recursive feedback loops, making resistance increasingly difficult as different forms of control become inseparable and mutually reinforcing. These mechanisms are already at work in contemporary society. From recommendation algorithms to surveillance systems, AI increasingly shapes not just what we see but how we think and behave. The difference between present reality and future possibilities is one of degree rather than kind. The theoretical framework developed here suggests both dangers and possibilities. While AI systems threaten to create a society of total algorithmic control, understanding their mechanisms also reveals potential points of resistance. The challenge lies in moving beyond both techno-utopianism and dystopian fatalism toward approaches that harness AI's reality-shaping power for human flourishing rather than algorithmic optimization and dividualization. Looking forward, research must focus on practical strategies for maintaining human agency within algorithmic nexus environments, such as implementing explainable AI systems, mandating algorithmic transparency and contestability, fostering digital literacy and critical algorithmic awareness, designing opt-out mechanisms, and incorporating human-in-the-loop decision architectures.

"...In that Empire, the Art of Cartography attained such Perfection that the map of a single Province occupied the entirety of a City, and the map of the Empire, the entirety of a Province. In time, those Unconscion-

able Maps no longer satisfied, and the Cartographers Guilds struck a Map of the Empire whose size was that of the Empire, and which coincided point for point with it. The following Generations, who were not so fond of the Study of Cartography as their Forebears had been, saw that that vast Map was Useless, and not without some Pitilessness was it, that they delivered it up to the Inclemencies of Sun and Winters. In the Deserts of the West, still today, there are Tattered Ruins of that Map, inhabited by Animals and Beggars; in all the Land there is no other Relic of the Disciplines of Geography.”

–Borges (1946, On Exactitude in Science)

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Declarations

Conflict of interest The authors declare no competing interests.

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